

GG-Lap: A Bayesian Algorithm for Robust Reconstruction of Sparse EEG Causal Brain Networks (Supplement)

I. CONFUSION MATRIX

To provide a more intuitive illustration of the causal connectivity recovery results across the different methods, we employed a confusion matrix $e = e_{kl} \in \mathbb{R}^{3 \times 3}$ to analyze the ground truth causal networks that were successfully reconstructed [1]:

$$e_{lk} = \sum_{i=1}^N \sum_{j=1}^N P(D_{ij} \in l | C_{ij} \in k); \quad l = 1, 2, 3; k = 1, 2, 3 \quad (1)$$

where C and D represent the predefined network and the recovered network. In the confusion matrix, different values of l and k represent the absence of connections, unidirectional connections, and bidirectional connections, respectively. Generally, a higher proportion of diagonal elements in the confusion matrix indicates a more accurate causal network reconstruction.

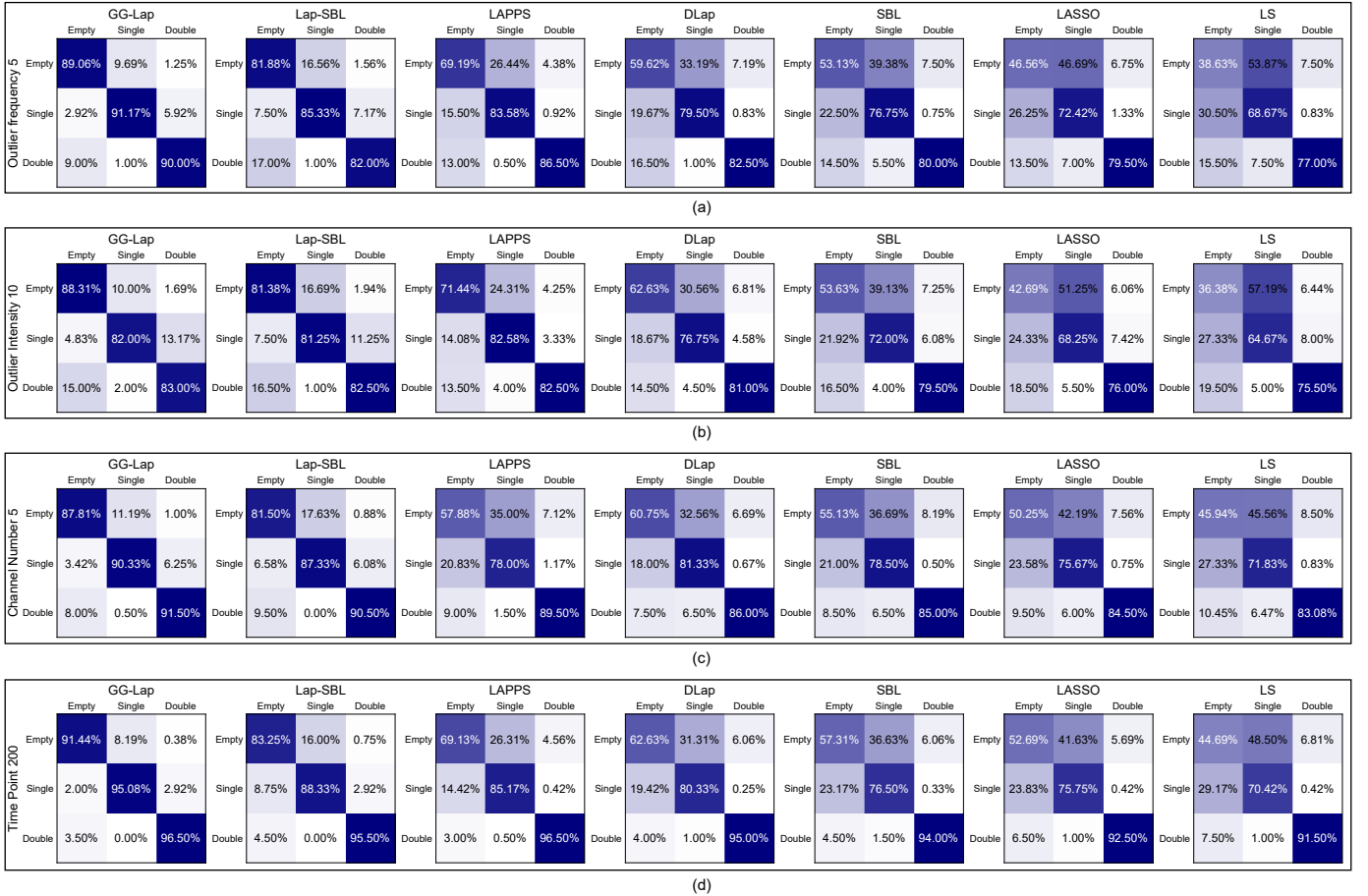


Fig. 1. Confusion matrices of GG-Lap, Lap-SBL, LAPPS, DLap, SBL, LASSO and LS under different outlier settings.

Fig. 1 highlights the topological differences in causal connections constructed by various methods. Clearly, across different conditions, the diagonal elements of the confusion matrices for GG-Lap show a more prominent presence compared to other algorithms. This suggests that GG-Lap more accurately infers inherent network topology, showcasing superior robustness in performance.

II. L_p -NORM AND GENERALIZED GAUSSIAN PRIOR

The solution obtained through L_p -norm regularization under the conventional regularization framework is equivalent to the Maximum A Posteriori (MAP) estimation with generalized Gaussian priors in Bayesian inference [2]. The following is the relevant mathematical derivation:

$$Y = AX + \varepsilon \quad (2)$$

where $\mathbf{Y} = [\mathbf{y}_{q+1}, \mathbf{y}_{q+2}, \dots, \mathbf{y}_T]^\top \in \mathbb{R}^{(T-q) \times N}$ represents the $T - q$ time points to be predicted. $\mathbf{A} = [\mathbf{A}_1^\top, \mathbf{A}_2^\top, \dots, \mathbf{A}_q^\top] \in \mathbb{R}^{(T-q) \times qN}$ is the design matrix for prediction composed based on the known data. $\mathbf{X} = [\mathbf{W}_1, \mathbf{W}_2, \dots, \mathbf{W}_q]^\top \in \mathbb{R}^{qN \times N}$ represents the parameters to be estimated for the MVAR model of the overall time series signal, and $\boldsymbol{\varepsilon} = [\boldsymbol{\varepsilon}_{q+1}, \boldsymbol{\varepsilon}_{q+2}, \dots, \boldsymbol{\varepsilon}_T]^\top \in \mathbb{R}^{(T-q) \times N}$ represents the overall noise residuals.

For the model prior, we assume that each row of $\mathbf{X}_{i,:} \in \mathbb{R}^{1 \times N}$ follows an independent generalized Gaussian distribution:

$$p(\mathbf{X}) \propto \exp \left\{ -\alpha \sum_{i=1}^{qN} \|\mathbf{X}_{i,:}\|_p^p \right\} \quad (3)$$

Taking negative log:

$$-\log p(\mathbf{X}) = \alpha \sum_{i=1}^{qN} \|\mathbf{X}_{i,:}\|_p^p + c_1 \quad (4)$$

Where c_1 is a constant.

For the model likelihood, assuming each column $\boldsymbol{\varepsilon}_{:,t} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$ is independent and identically distributed. Then:

$$p(\mathbf{Y}|\mathbf{X}) \propto \exp \left(-\frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_{\mathcal{F}}^2 \right) \quad (5)$$

Taking negative log:

$$-\log p(\mathbf{Y}|\mathbf{X}) = \frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_{\mathcal{F}}^2 + c_2 \quad (6)$$

Where c_2 is a constant.

For MAP estimation of $\mathbf{X}$, we aim to solve:

$$\hat{\mathbf{X}}_{\text{MAP}} = \arg \max_{\mathbf{X}} p(\mathbf{X}|\mathbf{Y}) \propto p(\mathbf{Y}|\mathbf{X}) \cdot p(\mathbf{X}) \quad (7)$$

Taking the negative log, this reduces to the minimization problem:

$$\hat{\mathbf{X}}_{\text{MAP}} = \arg \min_{\mathbf{X}} [-\log p(\mathbf{Y}|\mathbf{X}) - \log p(\mathbf{X})] \quad (8)$$

Substituting the likelihood term from (6) and the prior term from (4) into (8) gives the MAP optimization objective:

$$\hat{\mathbf{X}}_{\text{MAP}} = \arg \min_{\mathbf{X}} \left[\frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_{\mathcal{F}}^2 + \alpha \sum_{i=1}^{qN} \|\mathbf{X}_{i,:}\|_p^p \right] \quad (9)$$

Let $\lambda = 2\sigma^2\alpha$, then we obtain the standard L_p -norm regularization problem:

$$\hat{\mathbf{X}}_{\text{MAP}} = \arg \min_{\mathbf{X}} \left[\|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_{\mathcal{F}}^2 + \lambda \sum_{i=1}^{qN} \|\mathbf{X}_{i,:}\|_p^p \right] \quad (10)$$

III. INTERHEMISPHERIC LATERALIZATION INDEX

Additionally, we computed the interhemispheric lateralization index (ILI) [3] for each algorithm-derived brain network to quantify the degree of asymmetry in information flow between the left and right cerebral hemispheres in OpenBMI motor imagery task EEG dataset:

$$ILI = \frac{\sum_{i=1}^n \sum_{j=1}^m [A(l_i, r_j) - A(r_j, l_i)]}{\sum_{i=1}^n \sum_{j=1}^m [A(l_i, r_j) + A(r_j, l_i)]} \quad (11)$$

where $A(l_i, r_j)$ corresponds to the connectivity strength from electrode l_i in the left hemisphere to electrode r_j in the right hemisphere, and $A(r_j, l_i)$ corresponds to the connectivity strength from electrode r_j in the right hemisphere to electrode l_i in the left hemisphere. Positive ILI values signify that information transfer within the brain network is predominantly directed from the left cerebral hemisphere to the right cerebral hemisphere, while negative values signify the reverse directional pattern. Tables I and II present the ILI calculation results.

TABLE I
ILIs OF SKILLED MI PERFORMANCE SUBJECTS IN OPENBMI.

	LS	LASSO	SBL	DLap	LAPPS	Lap-SBL	GG-Lap
Left-hand MI	0.1362*	0.1854*	0.2306*	0.3939*	0.3402*	0.5564*	0.8414
right-hand MI	-0.1704*	-0.2317*	-0.2539*	-0.3880*	-0.4296*	-0.5118*	-0.6240

TABLE II
ILIS OF MI ILLITERACY SUBJECTS IN OPENBMI.

	LS	LASSO	SBL	DLap	LAPPS	Lap-SBL	GG-Lap
Left-hand MI	0.0829*	0.1867*	0.1413*	0.1978	0.2145*	0.2300*	0.3241
right-hand MI	-0.1248*	-0.2091*	-0.1858*	-0.3726	-0.3513*	-0.3852*	-0.4309

As shown in Table I, the ILI results further indicate that brain networks constructed by GG-Lap exhibit higher degrees of interhemispheric information flow asymmetry during motor imagery tasks. Table II reveal that lateralization effects of all algorithms decrease to some extent in brain networks of MI-illiterate subjects. Nevertheless, GG-Lap maintains more significant lateralization than comparison algorithms even in these challenging MI subjects, with ILI results showing more pronounced information flow asymmetry.

IV. DETAILED DESCRIPTION OF EPILEPTIC EEG DATASET

The epilepsy dataset used in our experiments is derived from the official Brainstorm software tutorial and has been extensively utilized in numerous EEG source imaging studies [4]. The data originates from a patient with focal epilepsy who has suffered from focal sensory and cognitive impairments, as well as secondary generalized seizures since the age of 8. Non-invasive telemetry revealed left frontal-central spikes, multi-spikes, and beta band activity bursts (with maximum amplitude at FC1 and Cz), particularly during sleep. Subsequently, the patient underwent invasive EEG examination to localize the epileptogenic zone and received a customized left frontal resection. The dataset has a sampling rate of 256 Hz with 29 EEG channels. Prior to spike averaging, the data underwent bandpass filtering at 0.5-80 Hz, followed by ICA artifact removal and average re-referencing. For spike averaging, a total of 58 spike segments were extracted, with each segment spanning [-300, +500] ms around the spike events, resulting in a final data length of only 800 ms for causal brain network computation.

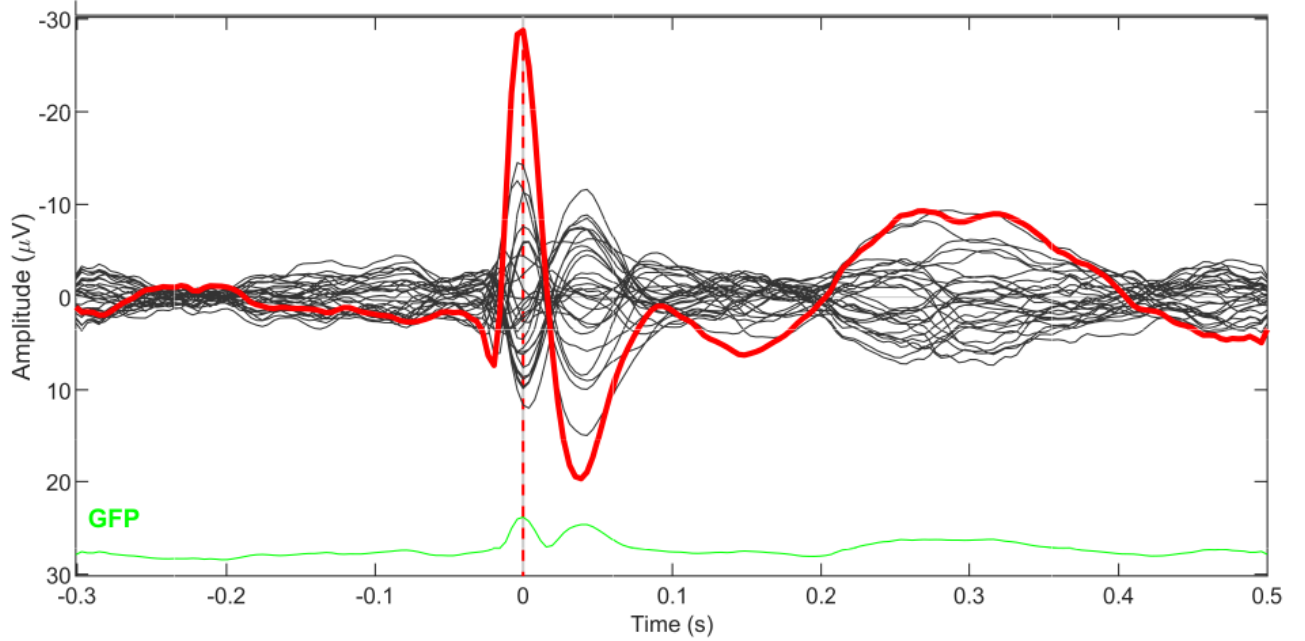


Fig. 2. Spike-averaged waveforms from epileptic EEG data.

In this study, we employed an averaging strategy across 58 epileptic spikes that were temporally aligned to their peak onset. Each individual spike segment can be conceptually decomposed into three components: (i) the spike-related neural response that is consistently time-locked across trials, (ii) slowly varying background activity that is not phase-locked to the spike (e.g., baseline drifts or ongoing rhythmic oscillations), and (iii) random broadband noise. By aligning the spike peaks and averaging across trials, the time-locked component is coherently preserved, whereas the non-phase-locked background fluctuations and stochastic noise tend to cancel out due to their random phase relationships.

From a theoretical standpoint, this procedure improves the signal-to-noise ratio (SNR) by approximately $\sqrt{N}$, where $N = 58$ yielding an SNR enhancement of about 7.6. As a consequence, the averaged spike waveform exhibits substantially attenuated slow drifts and background oscillatory variability, both of which are major sources of nonstationarity in single-trial data.

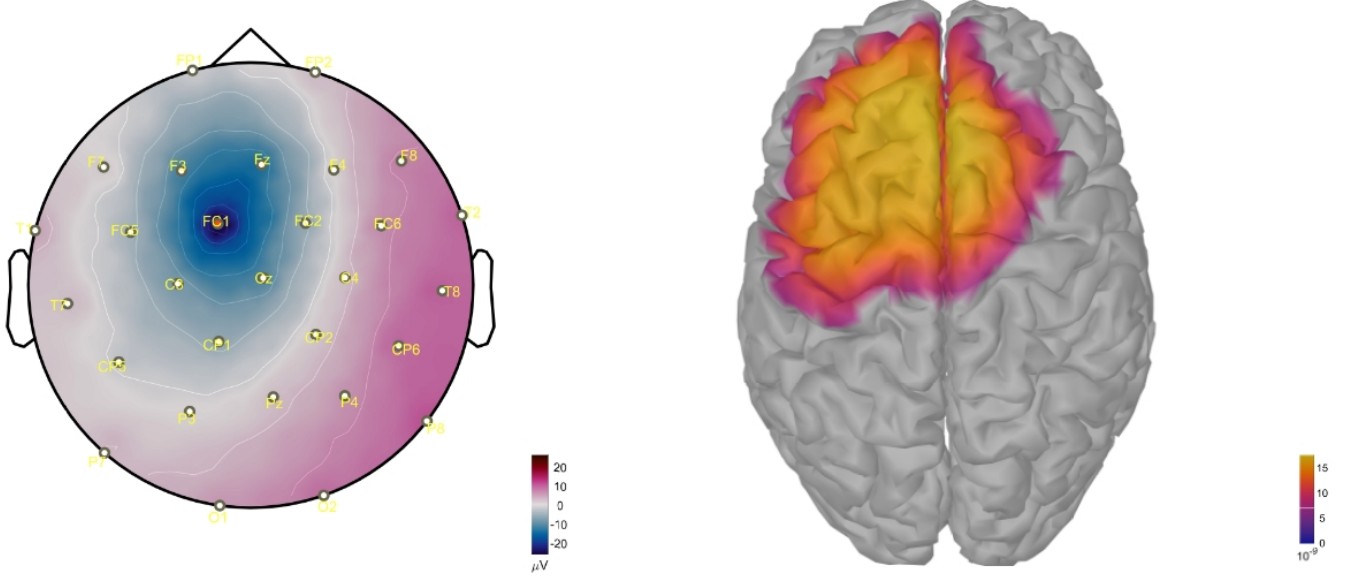


Fig. 3. Topographical map of epileptic EEG (left) and sLORETA algorithm source imaging (right).

Within the relatively short analysis window of 800 ms, the residual signals can therefore be reasonably approximated as weakly stationary, meaning that their mean and variance remain approximately stable over time. This averaging process not only preserves the event-related dynamics of the epileptic spike but also mitigates the influence of nonstationary components, providing a more robust basis for applying multivariate autoregressive modeling and Granger causality analysis. Consequently, we consider the spike-averaged epileptic EEG data segments to be appropriate for causal directed brain network analysis.

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